三角级数标准形式

$$T = \frac{2\pi}{\omega}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos n\omega x + b_n \sin n\omega x)$$

$$T = 2\pi$$

$$\frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

三角函数正交性

$$\int_{-\pi}^{\pi} \cos mx \cdot \sin nx \, dx = 0$$

$$\int_{-\pi}^{\pi} \cos mx \cdot \cos nx \, dx = \begin{cases} 0 & m \neq n \\ \pi & m = n \end{cases}$$

$$\int_{-\pi}^{\pi} \sin mx \cdot \sin nx \, dx = \begin{cases} 0 & m \neq n \\ \pi & m = n \end{cases}$$